
RangeViT: Towards Vision Transformers for 3D Semantic Segmentation in Autonomous Driving

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The Research Objective

Segmentation



Categorising different categories: cars, trees, pedestrians, etc. based on LiDAR images.

Unifying Network Architecture



Any advance in one domain benefits to both LiDAR point cloud and image challenges.

Transfer Power



Using 2D Image "Brains" to solve 3D spatial problems.

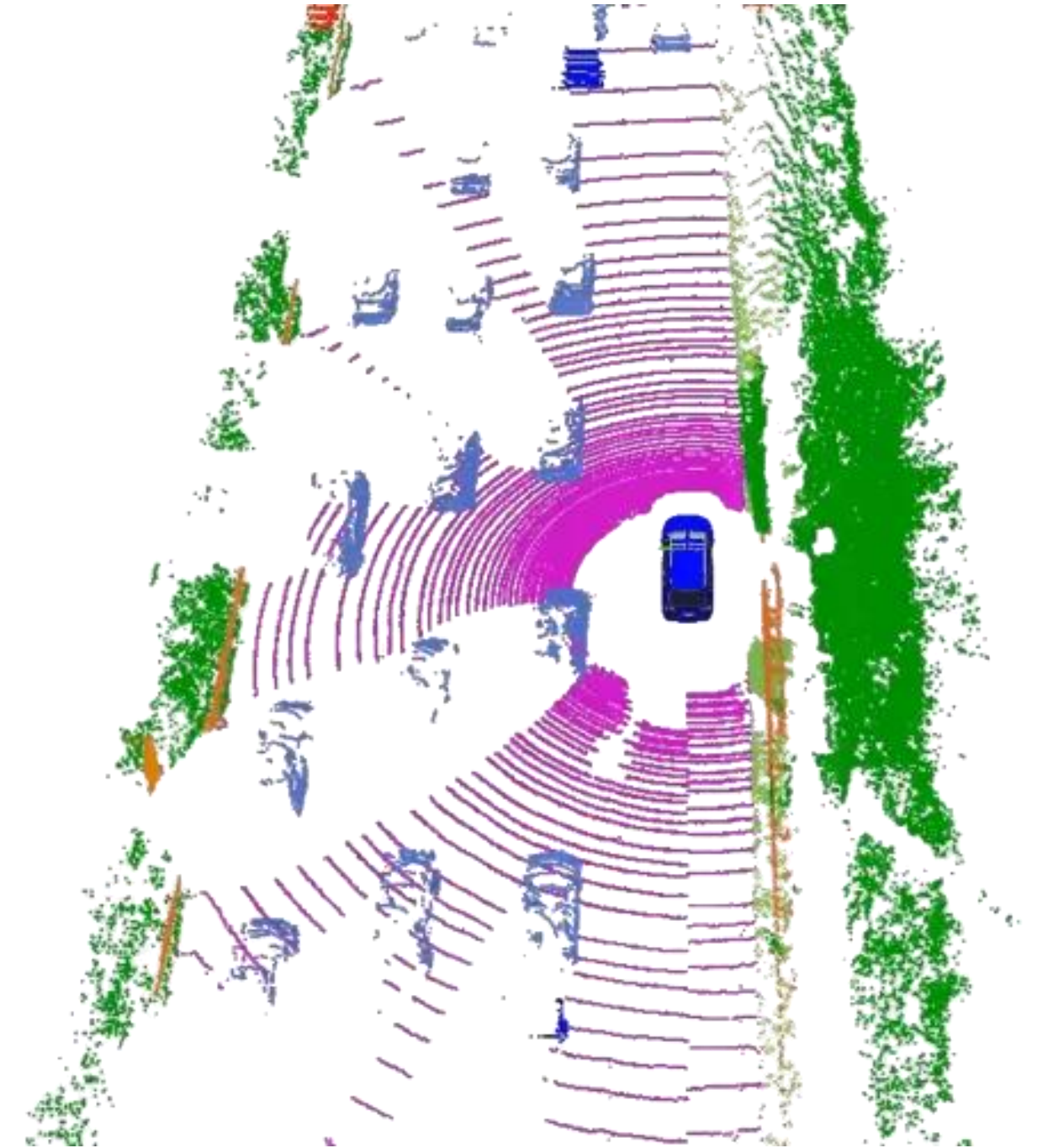
Understanding the Input: LiDAR

Raw Data: A "Point Cloud" of (x, y, z, i) coordinates.

- i : intensity
- (x, y, z) : Cartesian coordinates

The challenge: Data is sparse, unordered, and noisy as compared to Indoor LiDAR point clouds.

Datasets: SemanticKitti (64 laser beams, 19 classes),
NuScenes (32 laser beams, 16 classes).



Understanding the Input: LiDAR

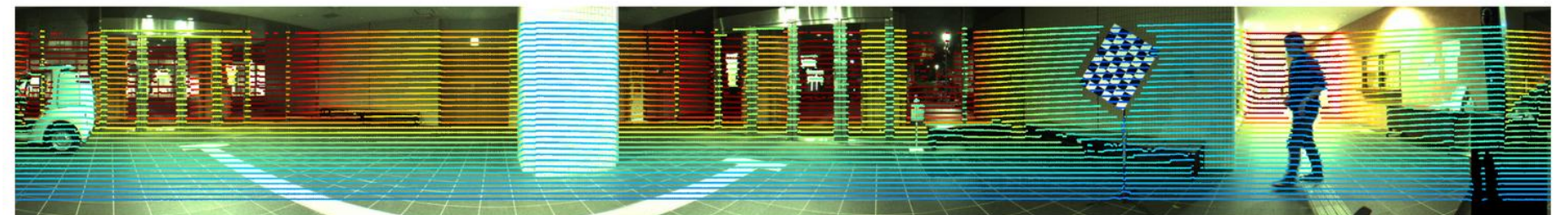
Literature Review:

- **3D/Voxel Methods:** High accuracy, but very slow. It's like trying to process a giant block of marble.
- **2D Projection Methods:** By flattening the world into a 2D picture, you lose the 3D geometry but it is faster.

Point cloud



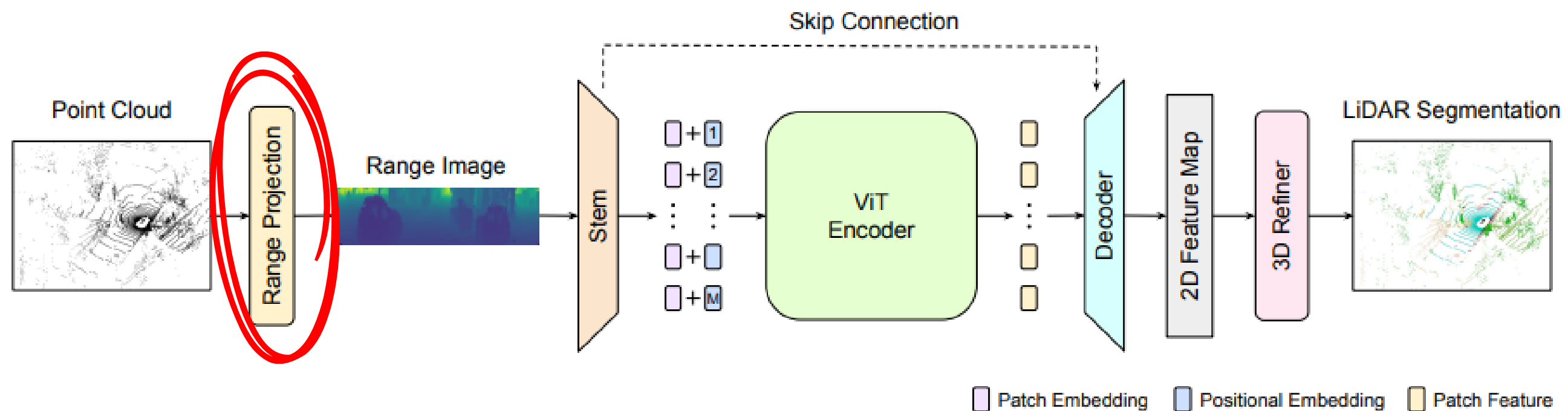
Voxel



Proposed Architecture

Range Projection:

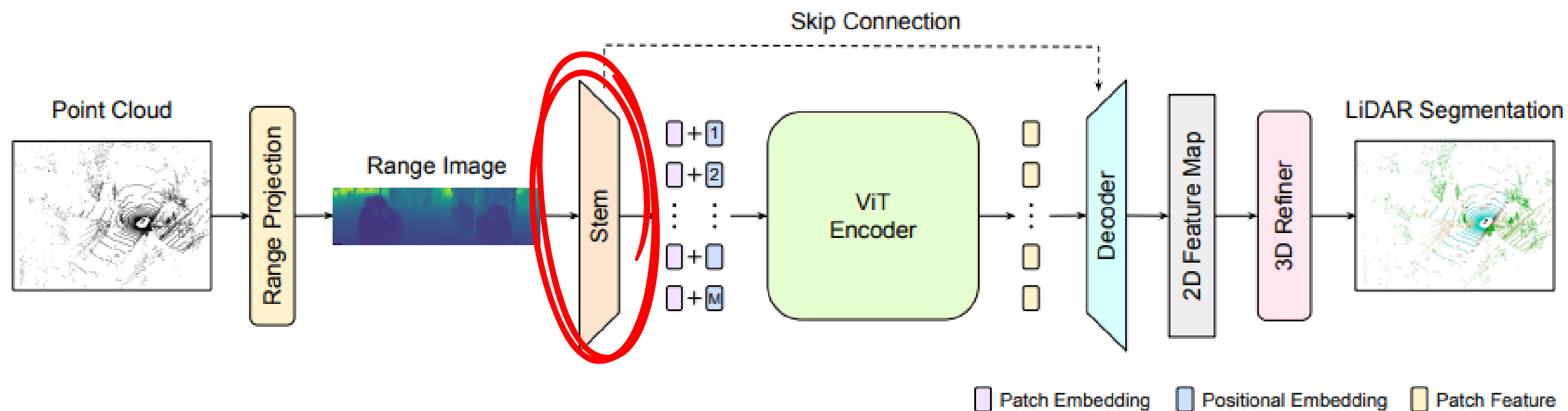
- **Input:** (x, y, z, i) raw 3D point cloud.
- **The Result:** A 5-channel image (r, x, y, z, i) . It looks like a long, panoramic photo where “r” represents the distance from the sensor.



Proposed Architecture

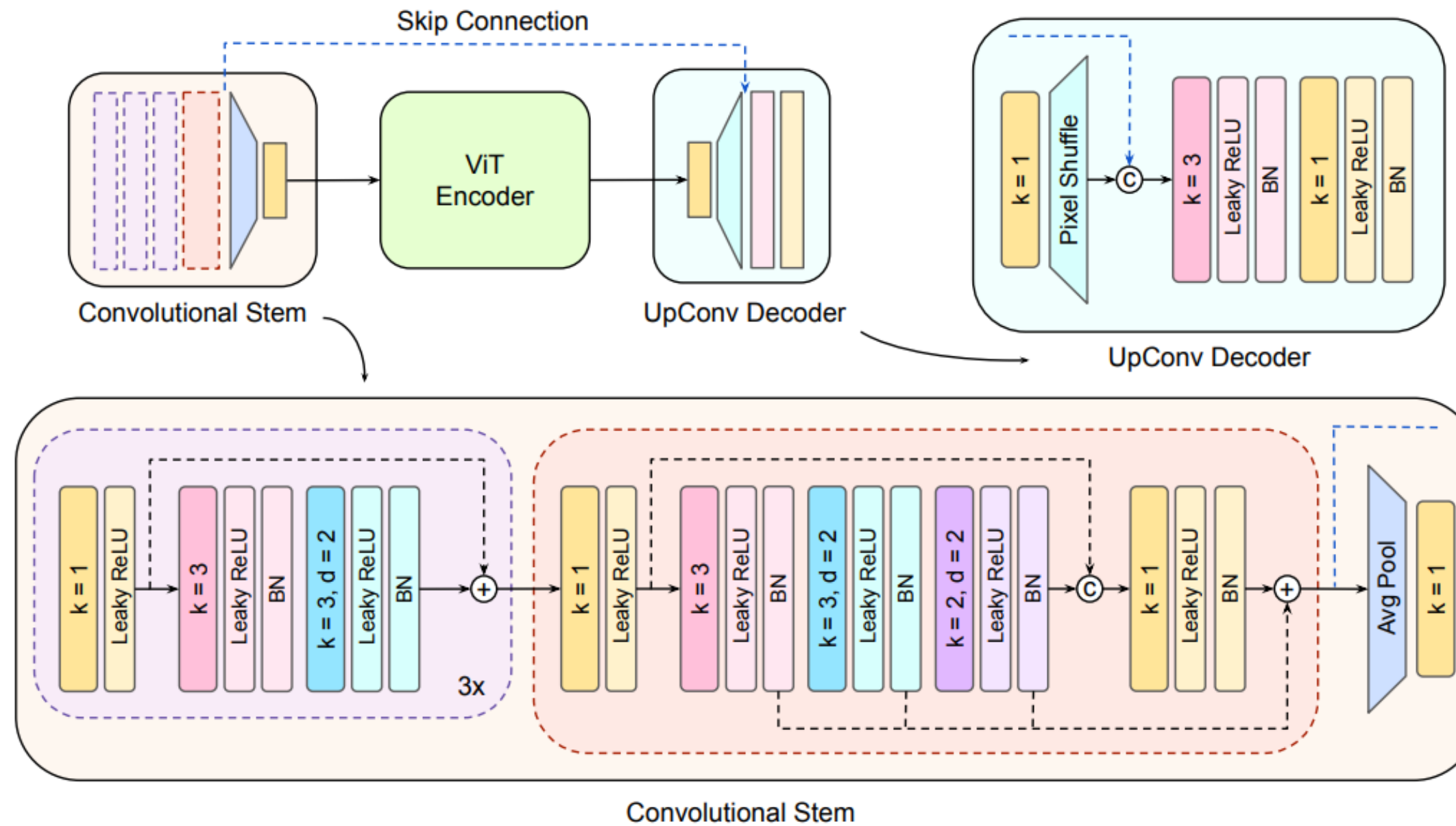
Convolutional Stem:

- **Problem:** ViTs can not understand the point cloud geometric patterns.
- **Conv:** captures local spatial relationships aka inductive bias.
- **Context module:** Captures short-range dependencies and produces pixel-wise features.
- **Average pooling layer:** Reduces dimensions to match as that of ViT input dimension



Proposed Architecture

Convolutional Stem:



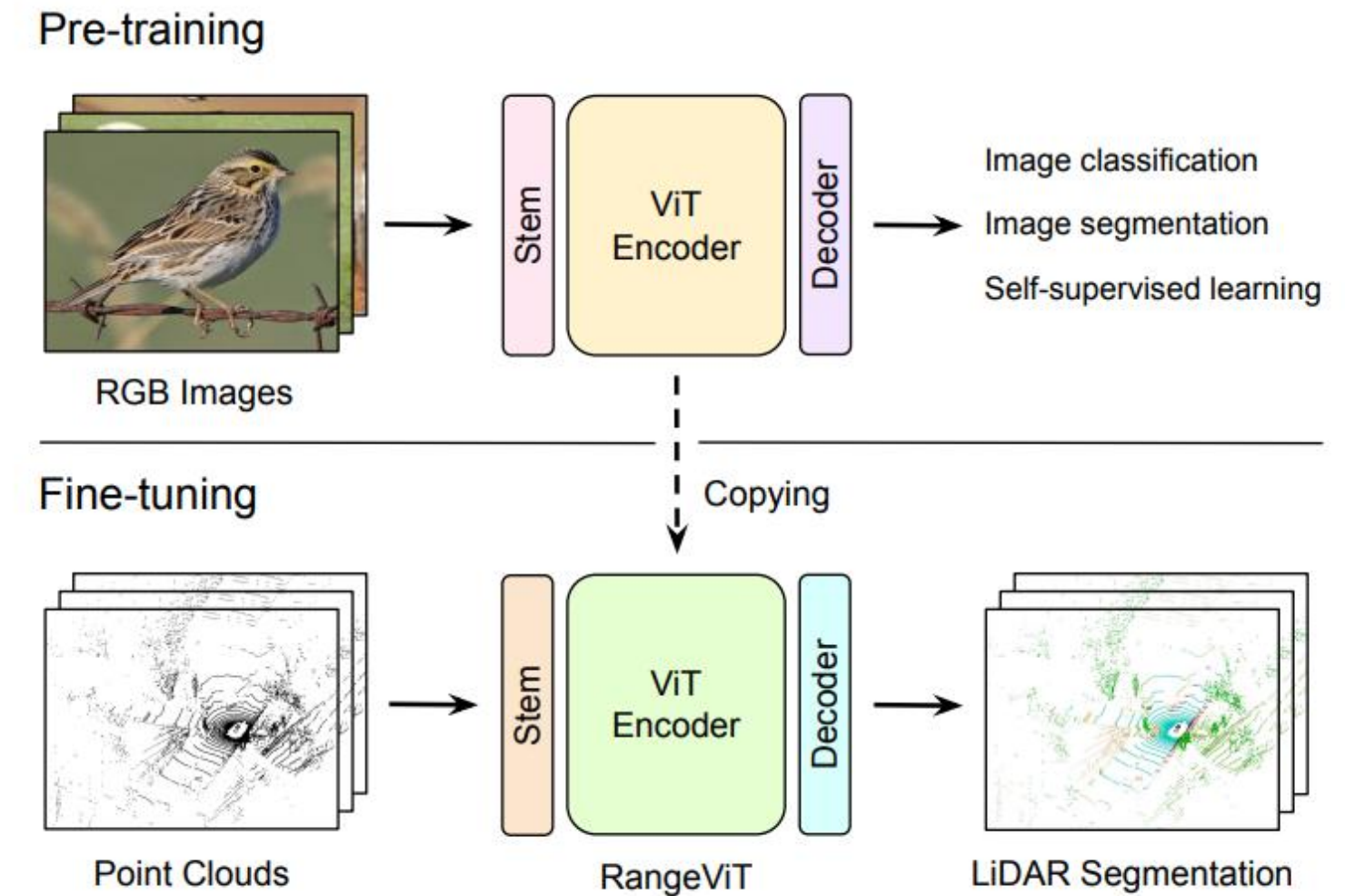
Proposed Architecture

ViT Encoder

Why ViT?

- Long-term dependencies
- Pretrained on RGB images
- Fine-tuned on CityScapes dataset
- “Attention” layers: handles relationships and contrasts.
- FFN: details and features of single-patch

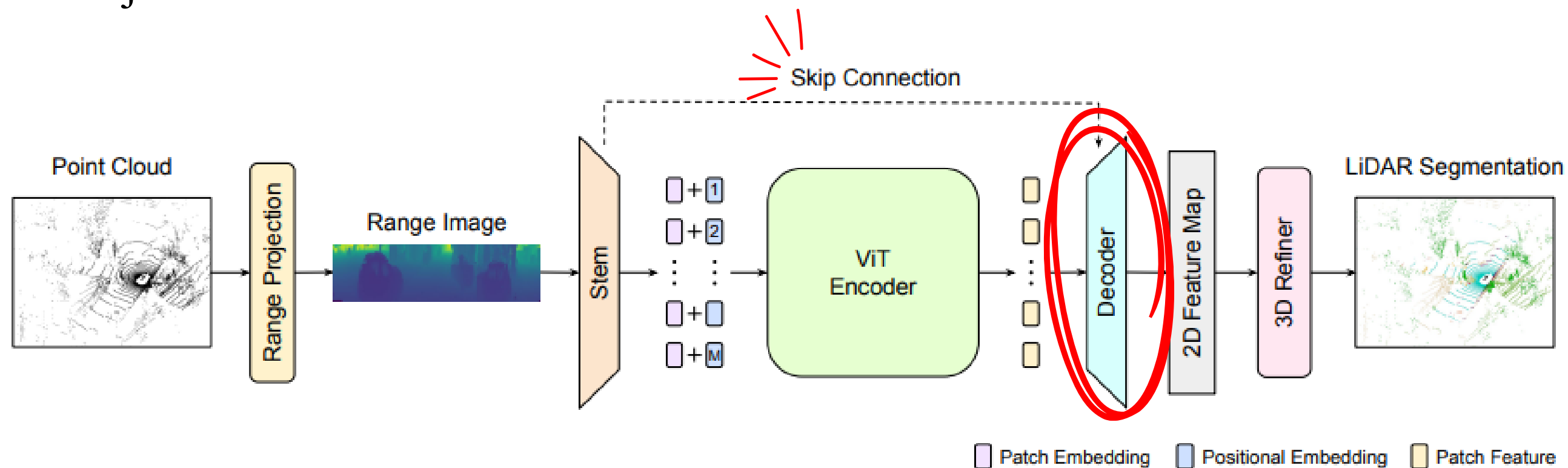
Therefore Global Attention (2x8 patches)



Proposed Architecture

Convolutional Decoder: (Sharpener)

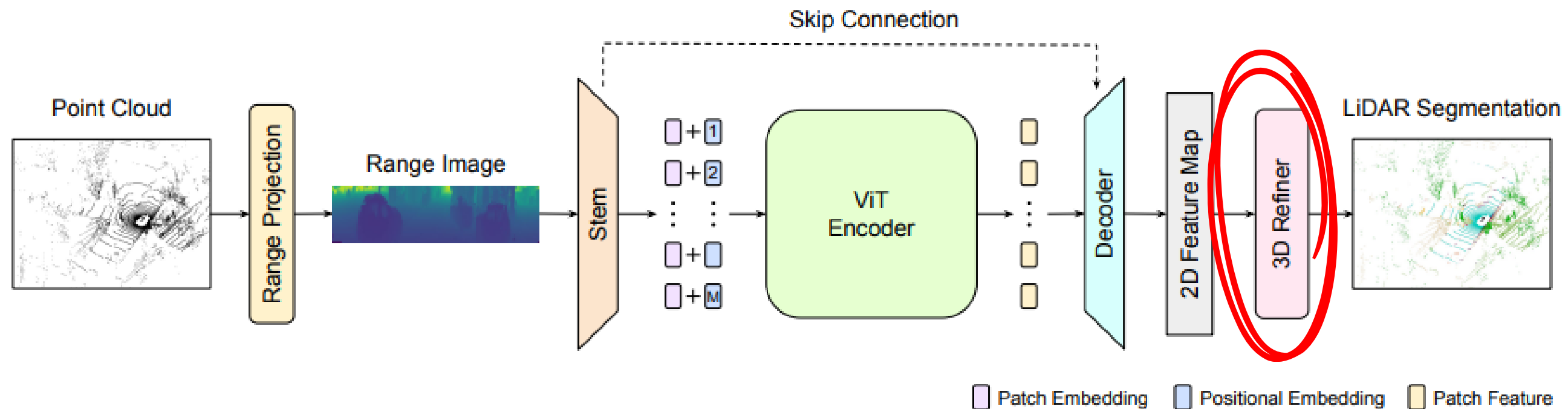
- **Problem:** Transformers are great at "thinking," but they produce low-resolution maps.
- **Decoder:** Uses Pixel Shuffle technique that rearranges deep data into sharp pixels.
- **Skip Connection:** Takes “sharp” details from the Conv Stem so we don't lose exact edges of the objects



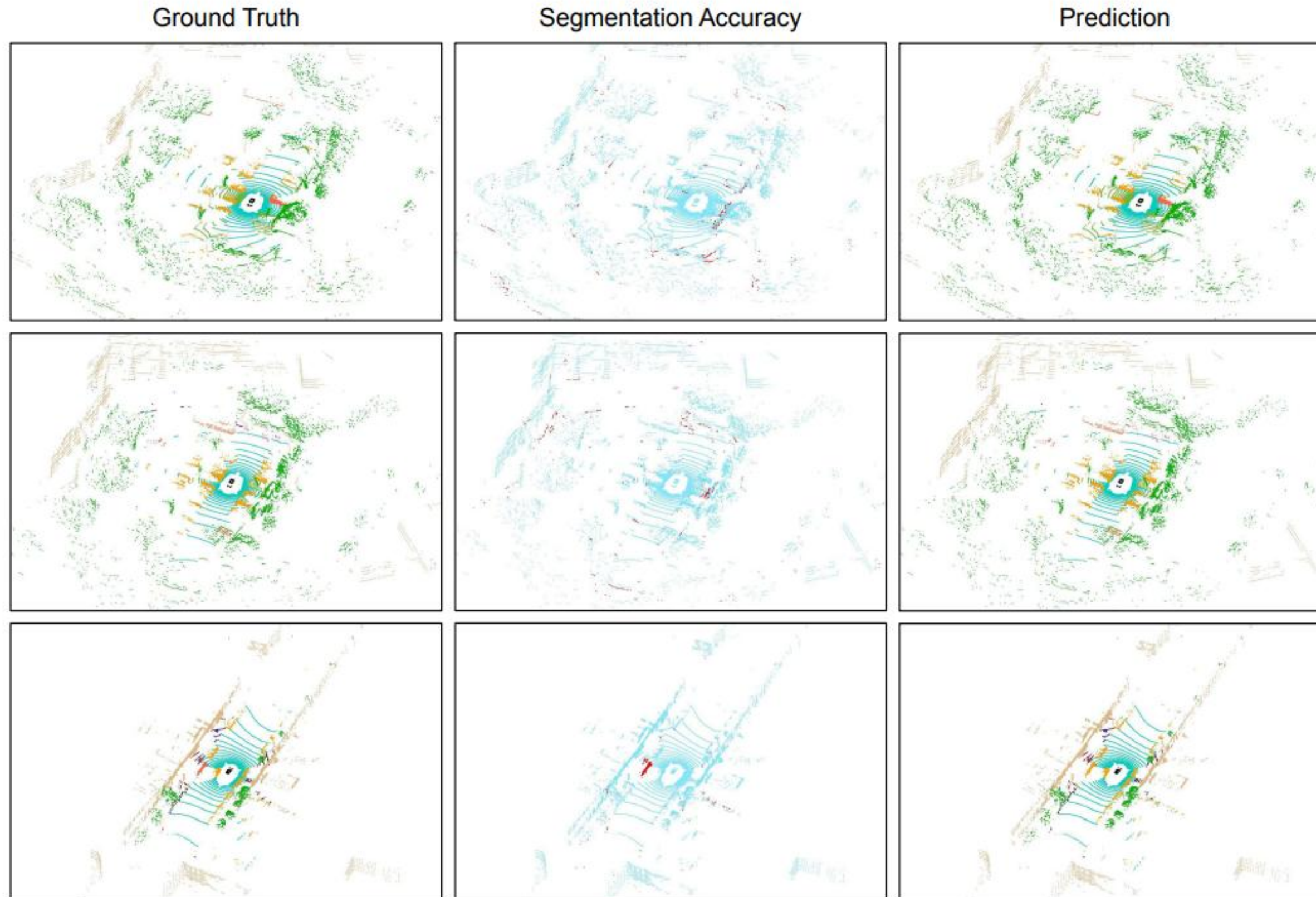
Proposed Architecture

3D Refiner:

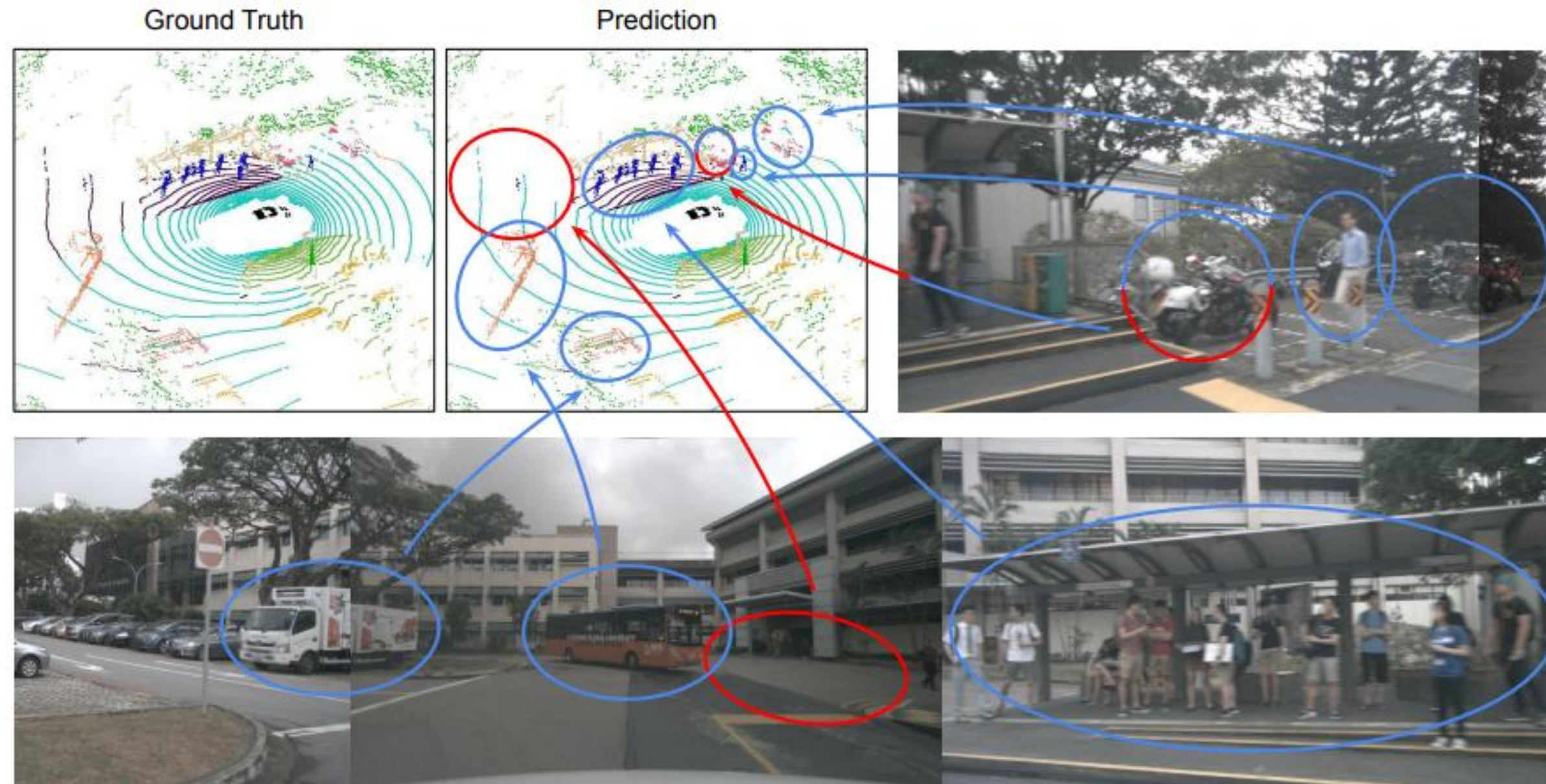
- **Problem:** 2D image, but we need 3D.
- **KPConv Layer:** The Kernel Point Convolution converts the 2D image to 3D and fixes overlaps that might occur during previous steps, and bilinear upsampling.
- **Final logits:** Probabilities of each pixel for each class.



Final Output



Final Output



Ground Truth Label Colours		Bicycle		Bus		Car		Sidewalk		Manmade		Terrain	
		Motorcycle		Truck		Pedestrian		Driveable Surface		Traffic Cone		Vegetation	

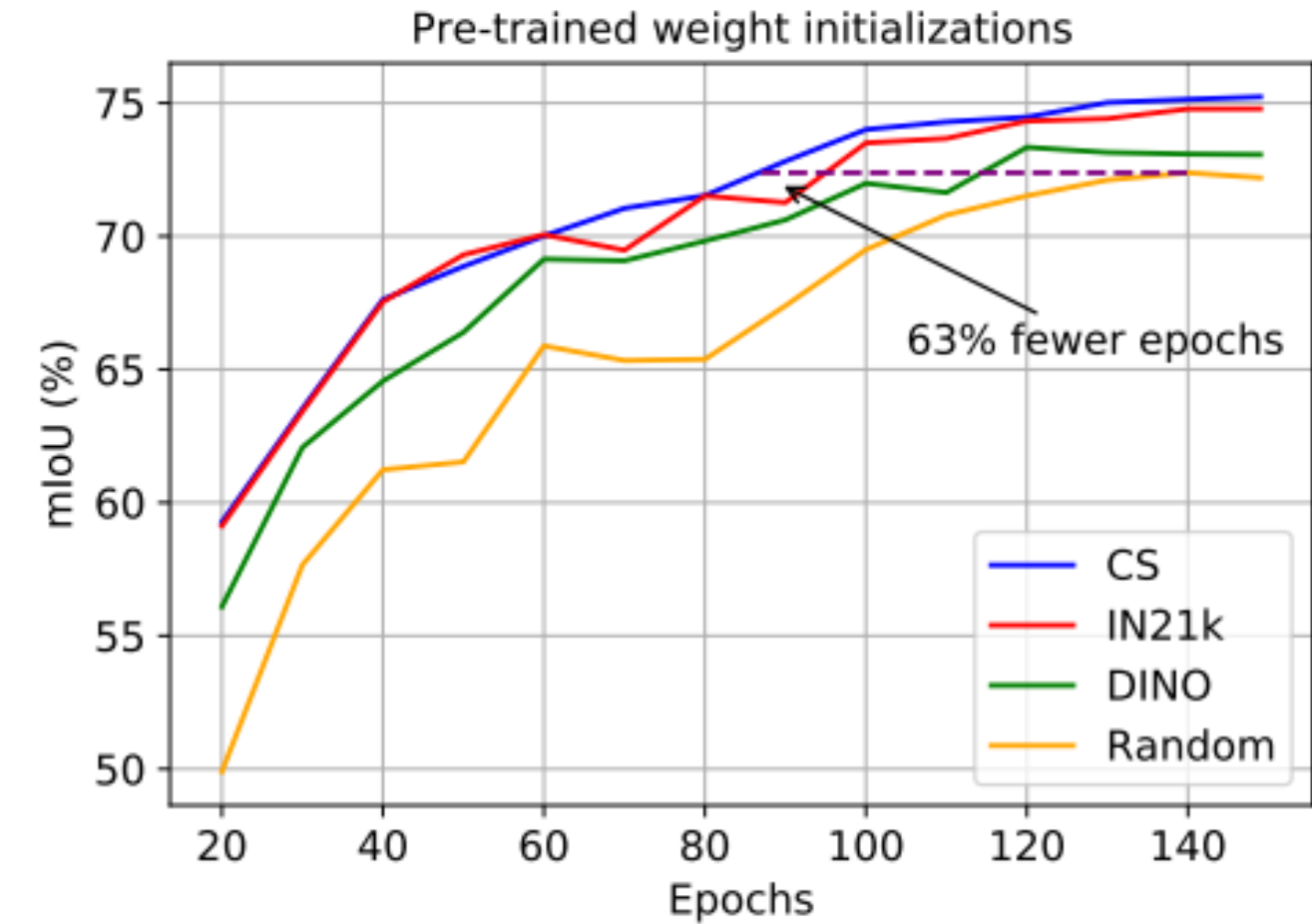
Results & Ablation Analysis

Model Ablation

Stem	Decoder	Refiner	mIoU	#Params
Linear	Linear		65.52	22.0M
Conv	Linear		69.82	22.8M
Conv	UpConv		73.83	24.6M
Conv	UpConv	✓	74.60	25.2M

Patch Size Ablation

Patch size	16×16	8×8	4×16	4×8	4×4	2×16	2×8
mIoU	68.45	72.04	72.72	73.30	73.70	73.88	75.21
#Tokens	49	193	193	385	769	385	769
Train time	$\times 1$	$\times 1.02$	$\times 1.02$	$\times 1.13$	$\times 1.43$	$\times 1.13$	$\times 1.43$



Cross Modal Transfer learning

Pre-training	Rand	DINO	IN21k	CS
mIoU	72.37	73.33	74.77	75.21

Conclusion:

Novelty of the research

- **ViTs are Ready for 3D:** Proven that standard Vision Transformer backbones (built for RGB images) can dominate LiDAR tasks with minimal architectural changes.
- **The "Cross-Modal" Win:** Using pre-trained weights from Cityscapes/ImageNet provides a "common sense" shortcut, leading to faster training and higher accuracy (+2.8 mIoU).
- **Best of Both Worlds:** RangeViT achieves the speed of 2D projection methods while approaching the accuracy of 3D voxel methods.
- **Efficiency & Scalability:** Through "Rectangular Tokenisation", the model handles massive LiDAR scans without memory explosion.
- **The Next Challenge:** While accurate, the model is still a "Black Box", leading into the necessity for Explainability (XAI).

Future work & Novelty:

Explainable AI

- Transformers are naturally more suitable for explainability than CNNs, because they use **attention mechanisms** that explicitly show which parts of the input influence the output.
- Despite this advantage, explainability for LiDAR transformers is still very underexplored.

What we will try to answer:

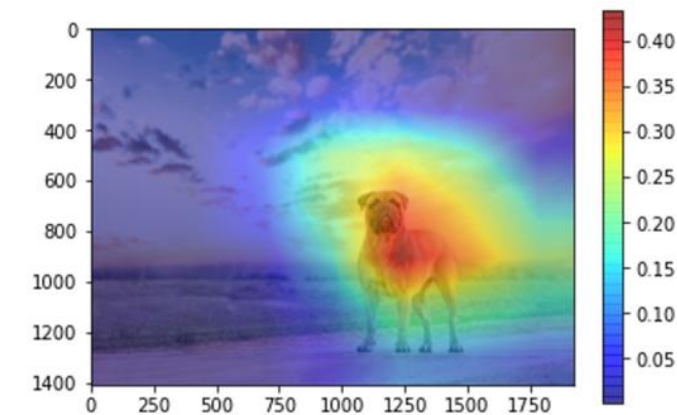
- Which part of LiDAR input influences each prediction?
- What kind of feature model focuses on?
- Where the model fails?

How will we answer:

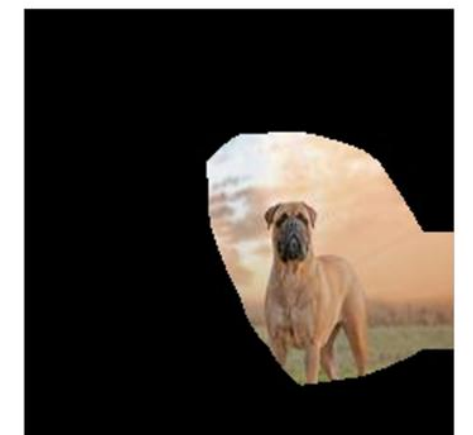
- Attention-based visualisation
- Identify top K important patches
- Study failure region



(a) Original image



(b) Attention map



(c) Visual explanation map

References

Pau Panareda Busto and Juergen Gall. Open set domain adaptation. In *2017 IEEE International Conference on Computer Vision (ICCV)*, pages 754–763, 2017. 2

Pau Panareda Busto, Ahsan Iqbal, and Juergen Gall. Open set domain adaptation for image and action recognition. *IEEE transactions on pattern analysis and machine intelligence*, 42(2):413–429, 2018. 2

Zhangjie Cao, Mingsheng Long, Jianmin Wang, and Michael I Jordan. Partial transfer learning with selective adversarial networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2724–2732, 2018. 2

Kaichao You, Mingsheng Long, Zhangjie Cao, Jianmin Wang, and Michael I Jordan. Universal domain adaptation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 2720–2729, 2019. 2

Jogendra Nath Kundu, Naveen Venkat, R Venkatesh Babu, et al. Universal source-free domain adaptation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 4544–4553, 2020. 2

Jogendra Nath Kundu, Naveen Venkat, Ambareesh Revanur, R Venkatesh Babu, et al. Towards inheritable models for open-set domain adaptation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 12376–12385, 2020. 2

Jogendra Nath Kundu, Naveen Venkat, Ambareesh Revanur, R Venkatesh Babu, et al. Towards inheritable models for open-set domain adaptation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 12376–12385, 2020. 2

Bo Fu, Zhangjie Cao, Mingsheng Long, and Jianmin Wang. Learning to detect open classes for universal domain adaptation. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XV 16*, pages 567–583. Springer, 2020. 2

Kuniaki Saito, Donghyun Kim, Stan Sclaroff, and Kate Saenko. Universal domain adaptation through self supervision. *Advances in neural information processing systems*, 33:16282–16292, 2020. 2

Kuniaki Saito and Kate Saenko. Ovanet: One-vs-all network for universal domain adaptation. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 9000–9009, 2021. 2

References

Dian Chen, Dequan Wang, Trevor Darrell, and Sayna Ebrahimi. Contrastive test-time adaptation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 295–305, June 2022. 1, 2, 3, 6, 13

Mario Döbler, Robert A Marsden, and Bin Yang. Robust mean teacher for continual and gradual test-time adaptation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 7704–7714, 2023. 2

Pascal Schlachter and Bin Yang. Comet: Contrastive mean teacher for online source-free universal domain adaptation. In *International Joint Conference on Neural Networks (IJCNN)*. IEEE, 2024. 1, 2, 3, 5, 6, 7, 8, 11, 13

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016. 8

Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschinot, Ce Liu, and Dilip Krishnan. Supervised contrastive learning. *Advances in neural information processing systems*, 33:18661–18673, 2020. 5

Dequan Wang, Evan Shelhamer, Shaoteng Liu, Bruno Olshausen, and Trevor Darrell. Tent: Fully test-time adaptation by entropy minimization. In *International Conference on Learning Representations*, 2021. 1, 2

Thank You!